Engineering Education in the Era of ChatGPT: *Promise and Pitfalls of Generative AI for Education*

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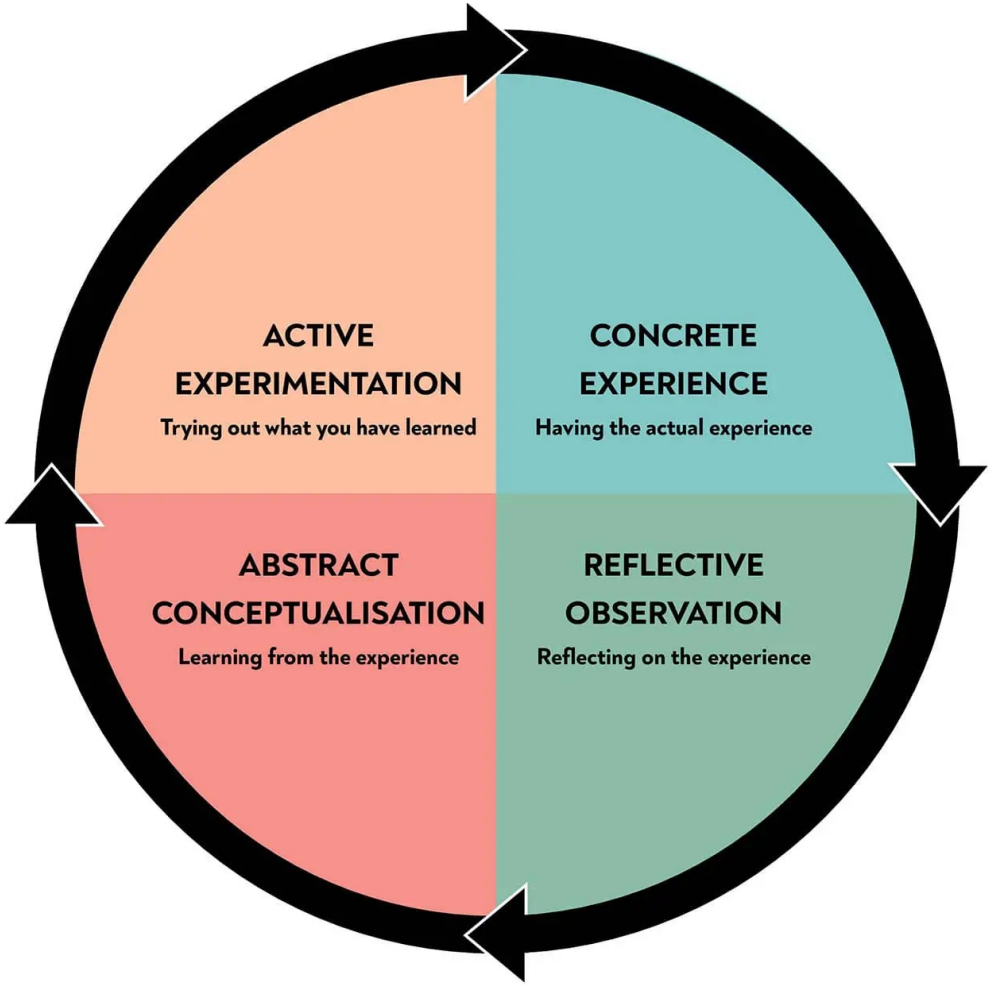
The Al-Ghazali Test?



You may use ChatGPT, but your learning is what you can do and retain when you don't have access to such a tool

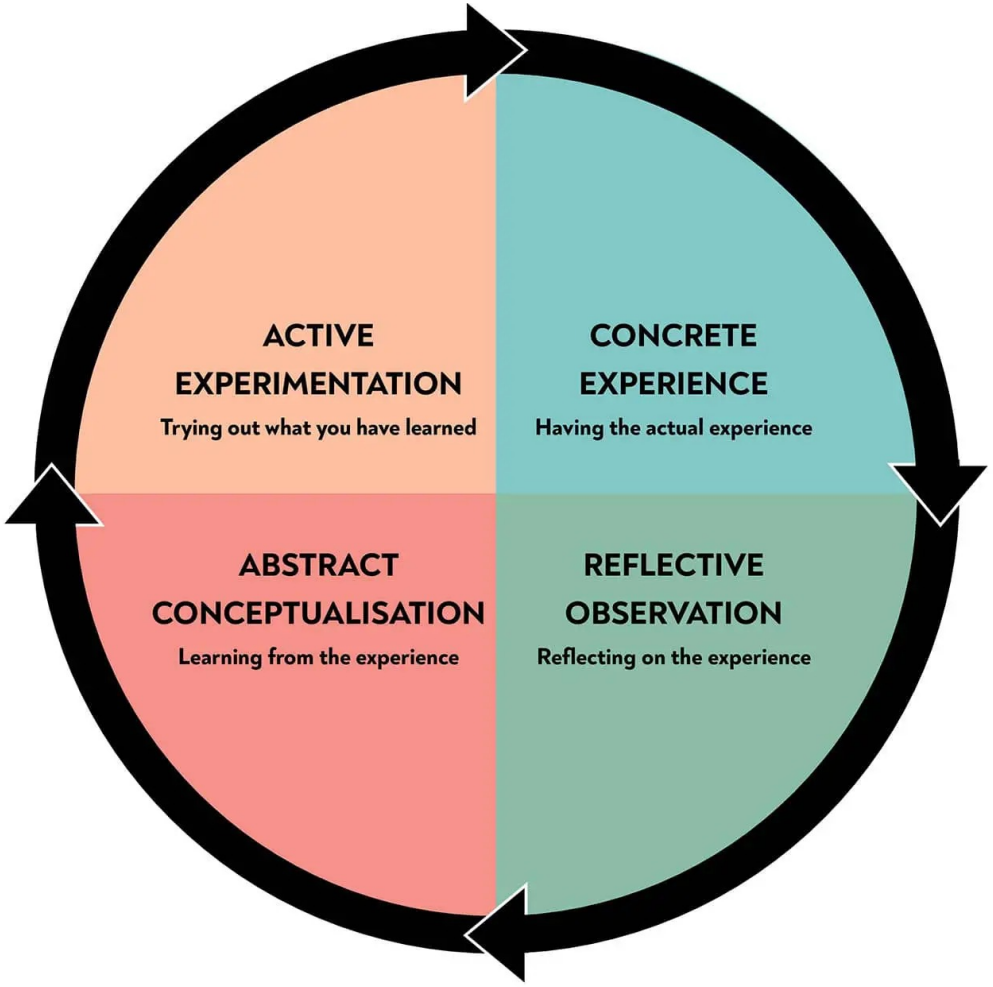
Opaqueness
Unpredictability
Hallucinations
Privacy violations
Harmful output (bias, mis/disinfo)
Inefficiency
…

Kolb's Learning Cycle



It has been proposed that the most profound and robust learning occurs if lesson activities follow the learning cycle first proposed by Kolb (1985), and later expanded on by Zull (2002) whereas the cycle begins with a tangible experience such as a designated video to watch or reading material to go through. This would be followed by reflective observation by the student on what they have learned through the assigned materials and afterwards the student reflects on abstract conceptualization of the material learned. Finally, this process ends with active experimentation of the use of material learned and the experience of receiving feedback from their instructor and dialog with other students through small groups or forum discussions (Kolb, 1985; Zull, 2002).

Enhancing Student's IQ (Inquiry Quotient)



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Metacognitive Strategies

In this method, the students write a few reflective sentences about the main points or takeaways that they received from the lesson that they just watched.

Reflective writing, when focused on the processes of learning, can help a student recognize areas of strength and weakness and then take conscious steps to improve with the low number of required resources to utilize this particular tool, reflective writing can be used in any classroom setting

Desirable difficulties and productive failures

Discussions where the primary objective is not necessarily in displaying a factually correct conclusion, but rather in fostering complex cognitive processes that hone one's critical thinking skills.

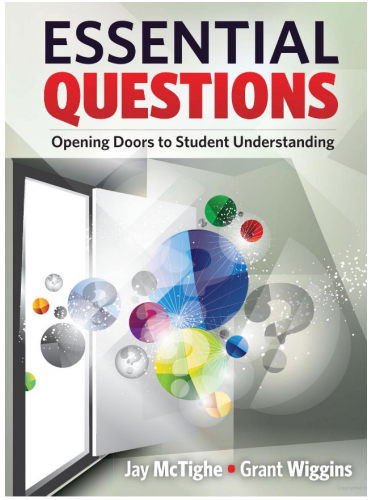
Essential questions and desired understanding

Understandings	**Essential Questions**
• The geography, climate, and natural resources of a region influence the economy and lifestyle of the people living there.	• How does *where* you live influence *how* you live?
• Statistical analysis and data display often reveal patterns. Patterns enable prediction.	• What will happen next? How sure are you?
• People have different dietary needs based on age, activity level, weight, and various health considerations.	• How can a diet that is "healthy" for one person be unhealthy for another?
• Dance is a language of shape, space, timing, and energy that can communicate ideas and feelings.	• How can motion express emotion?

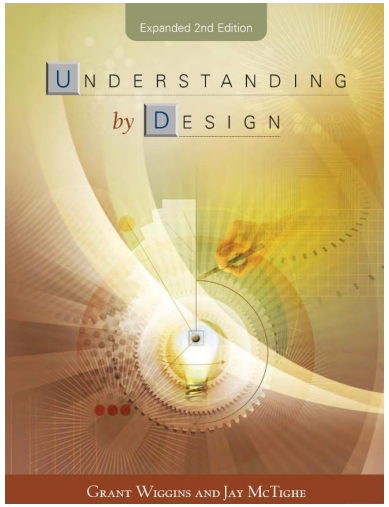
Essential questions (moving beyond factual/rote questions)

Essential Questions	Not Essential Questions
• How do the arts shape, as well as reflect, a culture?	• What common artistic symbols were used by the Incas and the Mayans?
• What do effective problem solvers do when they get stuck?	• What steps did you follow to get your answer?
• How strong is the scientific evidence?	• What is a variable in scientific investigations?
• Is there ever a "just" war?	• What key event sparked World War I?
• How can I sound more like a native speaker?	• What are common Spanish colloquialisms?
• Who is a true friend?	• Who is Maggie's best friend in the story?

What counts as an essential question



1. Is *open-ended;* that is, it typically will not have a single, final, and correct answer.
2. Is *thought-provoking* and *intellectually engaging*, often sparking discussion and debate.
3. Calls for *higher-order thinking*, such as analysis, inference, evaluation, prediction. It cannot be effectively answered by recall alone.
4. Points toward *important, transferable ideas* within (and sometimes across) disciplines.
5. Raises *additional questions* and sparks further inquiry.
6. Requires *support* and *justification*, not just an answer.
7. *Recurs* over time; that is, the question can and should be revisited again and again.

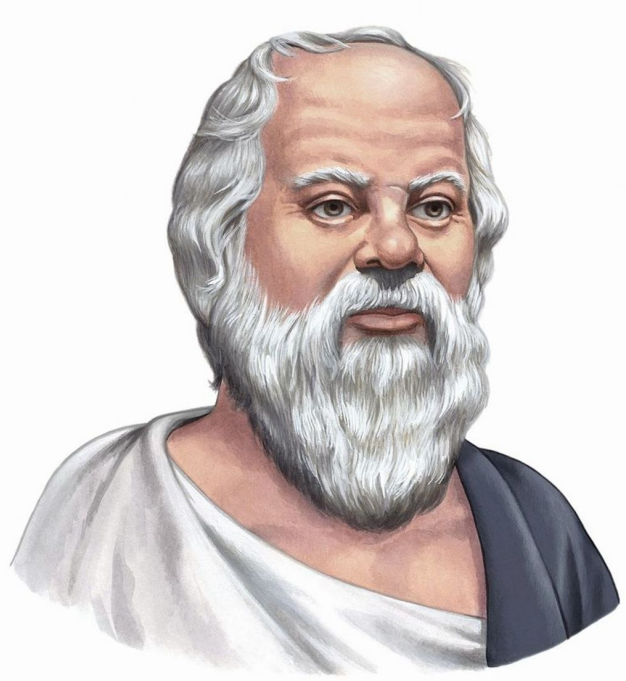


Content be unpacked to identify long-term transfer goals and desired understandings.

How to be in Goldilock's Zone

Desirable difficulties and productive failures

SOCRATES AND THE SOPHISTS



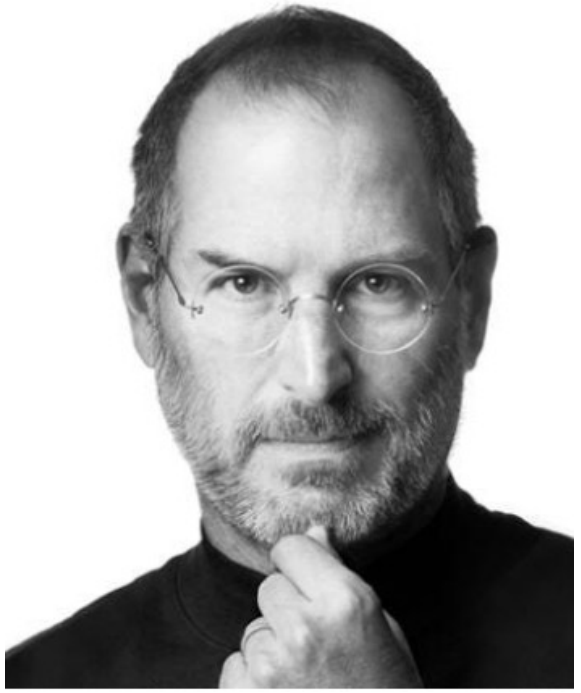
Appearences can be deceiving

AL-GHAZALI AND THE ROBBERS



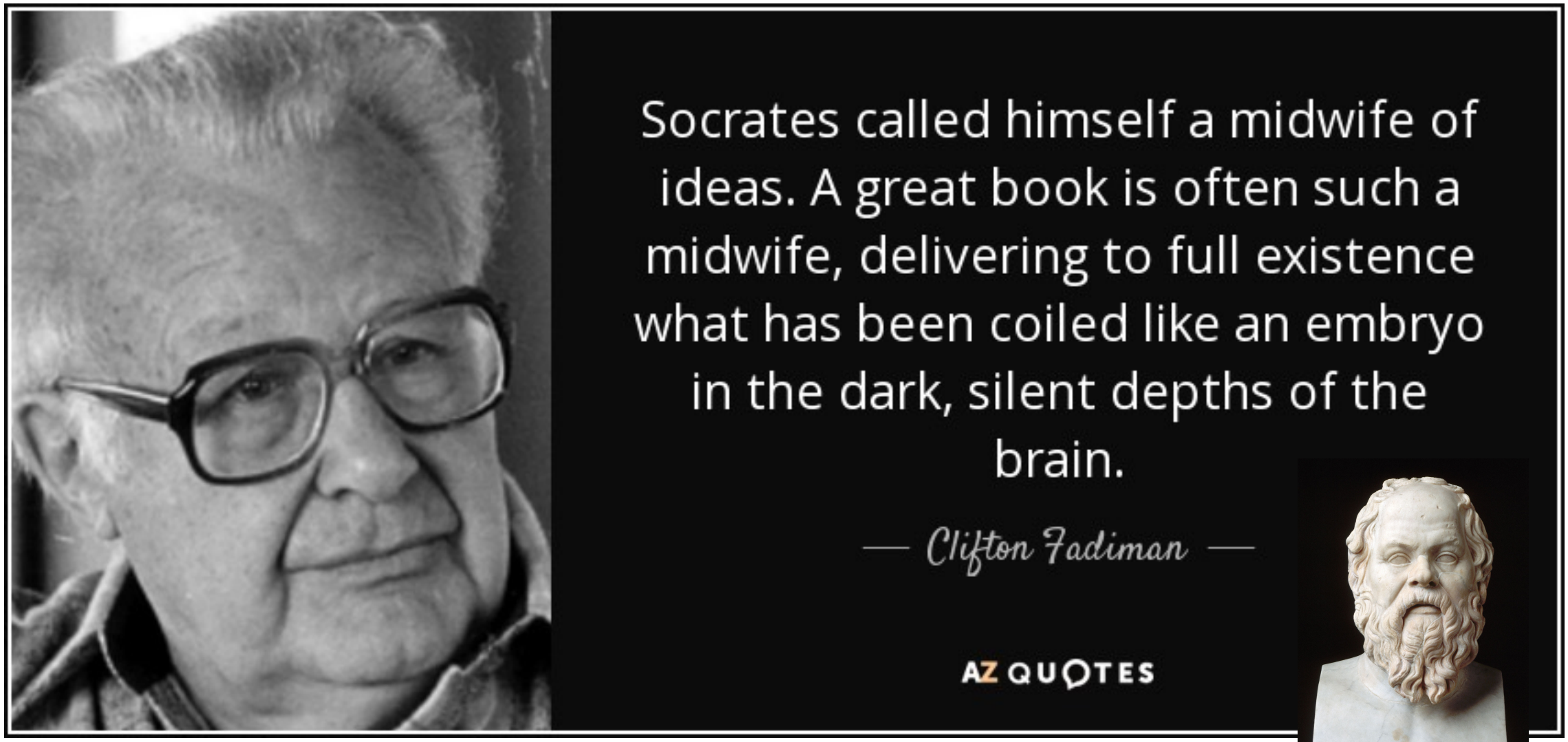
Education and knowledge is inside you

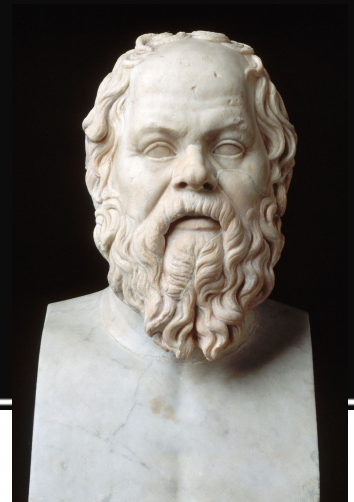
STEVE JOBS AND TYPOGRAPHY



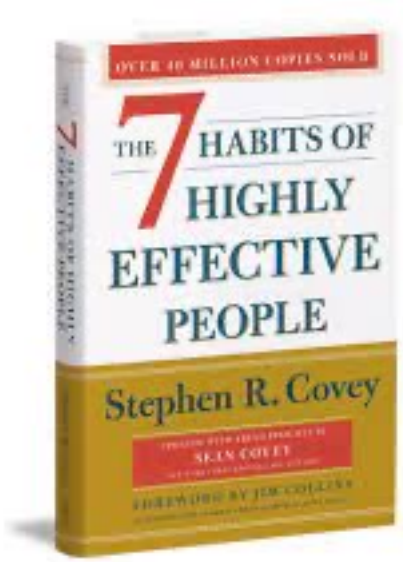
Effort spent learning is not wasted

Mid Wife of Understanding





Law of the Harvest: Beware of Shortcuts





Learning pathologies

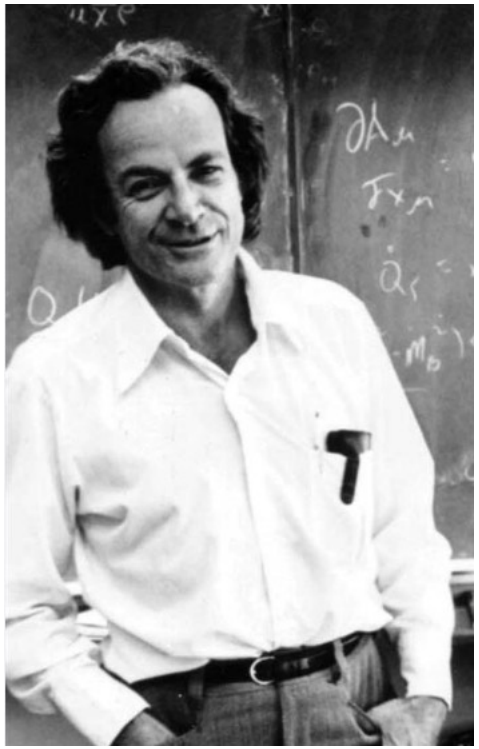
Cramming

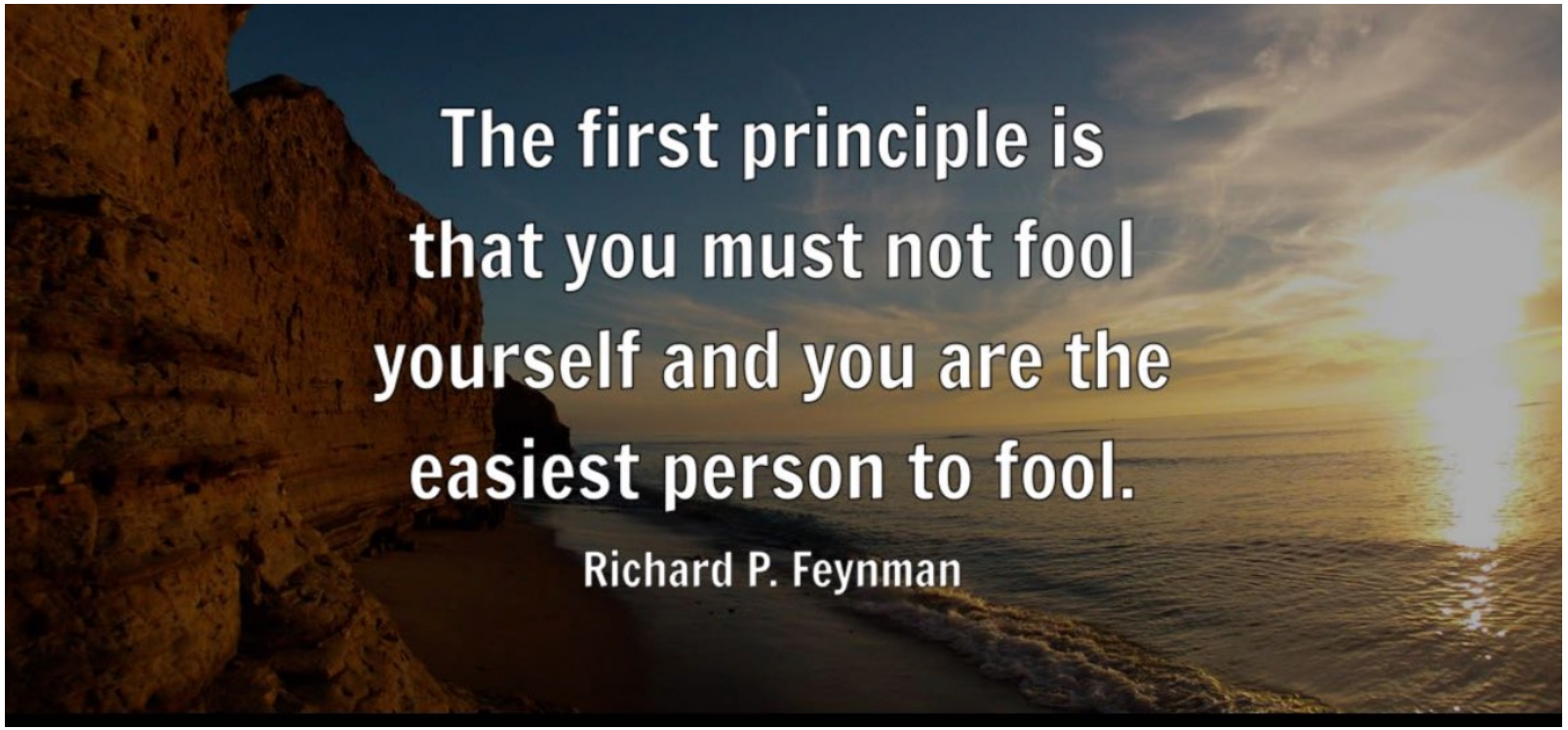
Gaming

Bribing

Critical Thinking Skills: More Important Than Ever

Cognitive Biases. Confirmation Bias.

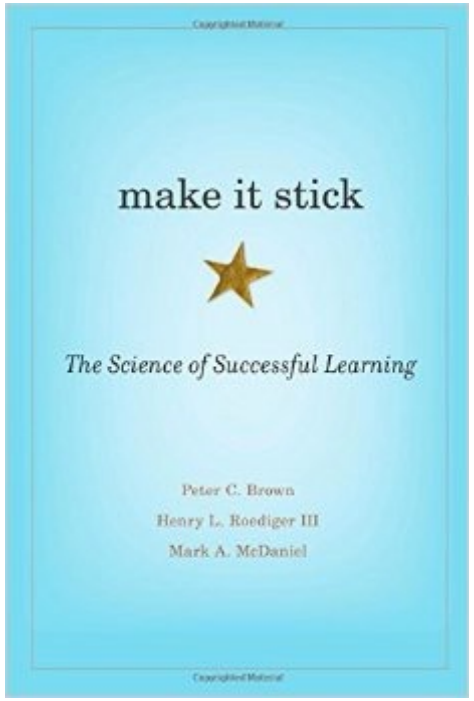




(Some) difficulties are desirable

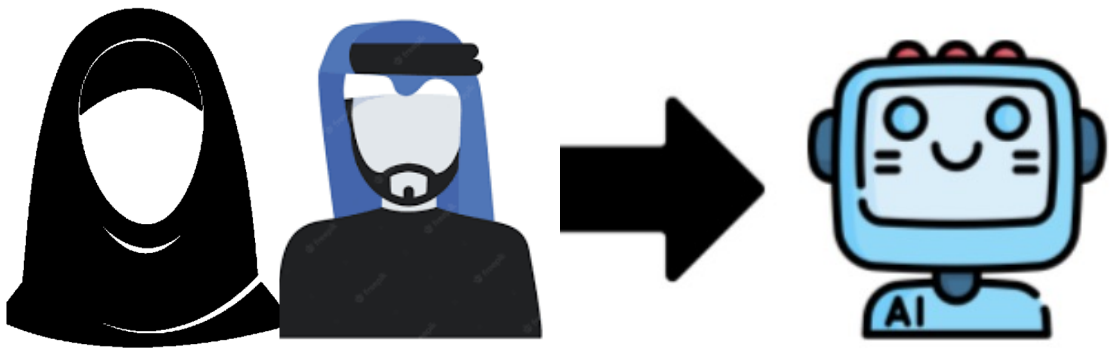
"One of the most striking research findings is the power of active retrieval—testing—to strengthen memory, and that **the more effortful the retrieval, the stronger the benefit**."

"by spacing or interleaving the practice, retrieval is harder, your performance is less accomplished, and **you feel let down, but your learning is deeper and you will retrieve it more easily in the future**."



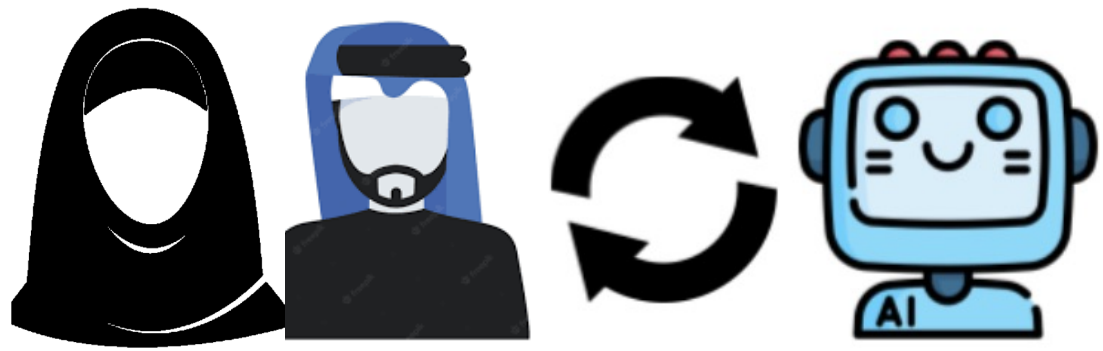
Automation vs. Augmentation

Automation



Replace human intelligence with artificial intelligence.

Augmentation



Augment human intelligence with artificial intelligence.

system traps and opportunities

Shifting the Burden *archetype occurs when a quick fix or easy solution is applied to a problem, rather than addressing the underlying root cause.*

Addiction, in the case of systems dynamics, is when agents become addicted to external forces to maintain the system

It's worth going through the withdrawal to get back to an unaddicted state, but it is far preferable to avoid addiction in the first place.

The problem can be avoided up front by intervening in such a way as *to strengthen the ability of the system to shoulder its own burdens*.



Learning results from what the student does and thinks, and only from what the student does and thinks. The teacher can advance learning only by influencing the student to learn.

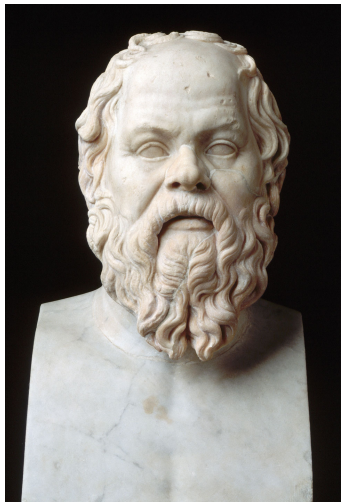
Herbert Simon

quotefancy

The Socratic Method: "The Midwifery of Ideas"

Socrates' educational philosophy can be described as a process of helping individuals give birth to their own ideas and knowledge, rather than simply imparting knowledge to them. This philosophy is often referred to as the Socratic method, which involves a series of questioning and dialogue in which the teacher helps the student to uncover their own thoughts and ideas.

Focus on covering rather than uncovering!



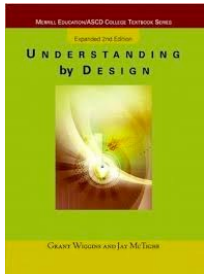
What are authentic skills?

"The most important method of education ... always has consisted of that in which the pupil was urged to actual performance."—Albert Einstein.



A performance task, problem, or project, is authentic if it:

(1) is realistic;
(2) requires judgment and innovation;
(3) engages the learner in "doing" the subject;
(4) replicates or simulates real-world contexts;
(5) assesses a learner's ability to deal with an open-ended complex task by using a repertoire of knowledge and skills;
(6) does not put artificial constraints on opportunities to practice, consult resources, obtain feedback, and refine performance.





We can change our assessments in two ways

You leave behind breadcrumbs using tools



Morally similar situations



Assessment that are hard to cheat again
Assessments that accommodate and facilitate human+AI collaboration

New ways of assessment are needed

Ask for things that cannot be (end-to-end) automated

Oral Assessment

Video Presentation

Progressive Assessment
(for newer more authentic tasks)

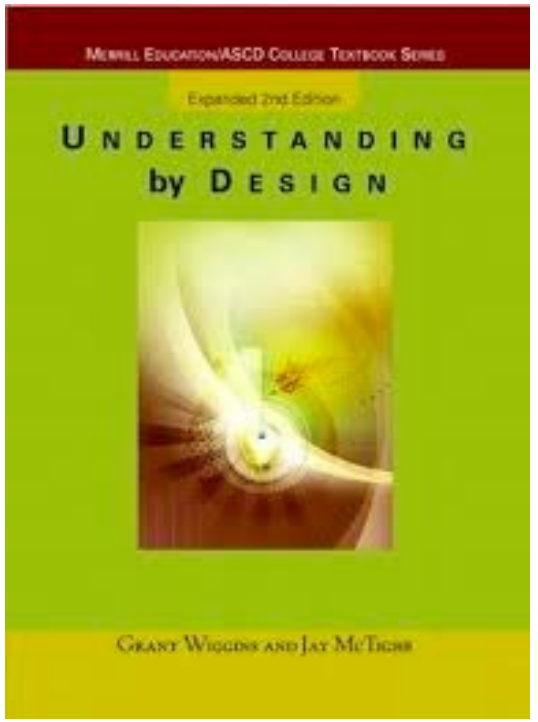
Oral Assessment

Moving towards authentic assessment

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Develop "Understandings" of "Big Ideas"

"…no skill can be integrated into a powerful repertoire unless the learner understands the big ideas related to using the skill wisely."
—Wiggins and McTighe.



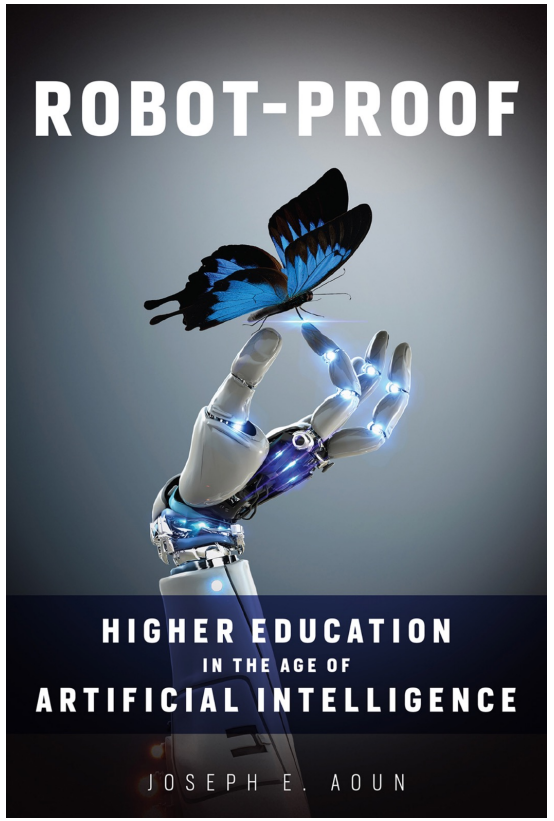
Share an "AI Tools Fair Usage Policy"

Your policy should include these considerations:

- Under what circumstances AI use is permitted or forbidden
- How students should cite or credit AI
- A warning about the technology's tendency toward hallucination (i.e., deceptive data) and clear rules regarding students' accountability for AI output
- A notice about using AI ethically and responsibly
- Discussion of the need to use AI as a tool to learn, not just to produce content

Experiential Learning

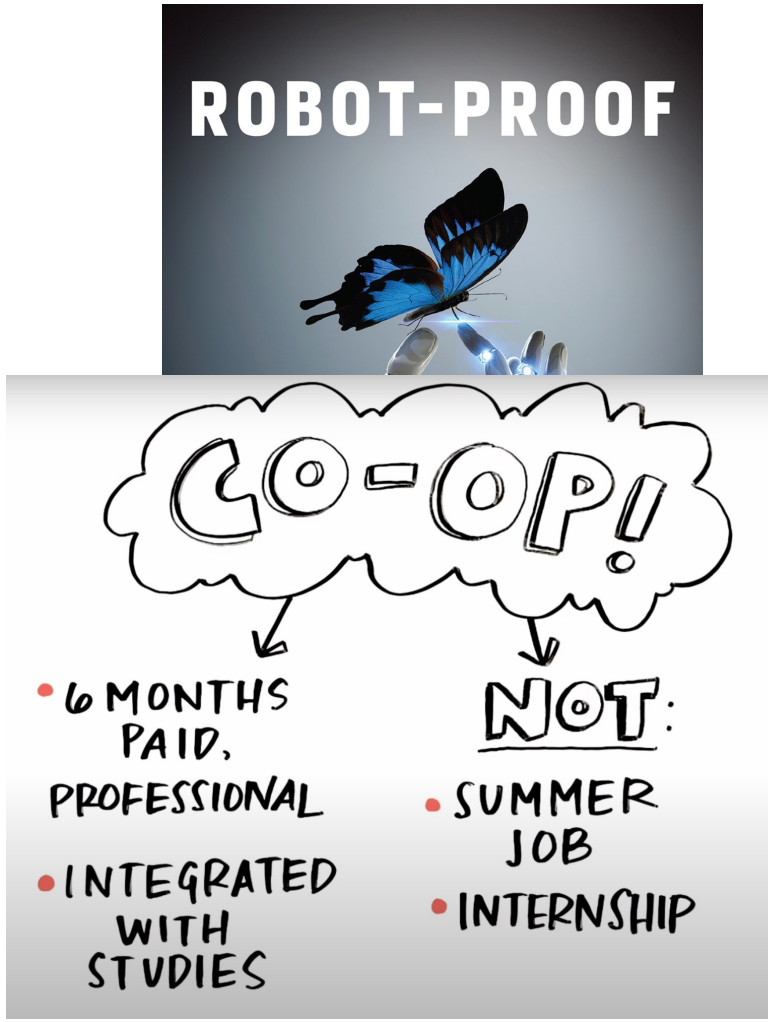
Typically, students engage in experiential learning through internships, co-ops, work-study jobs, global experiences, and original research opportunities.



Meaning making is what humans do …

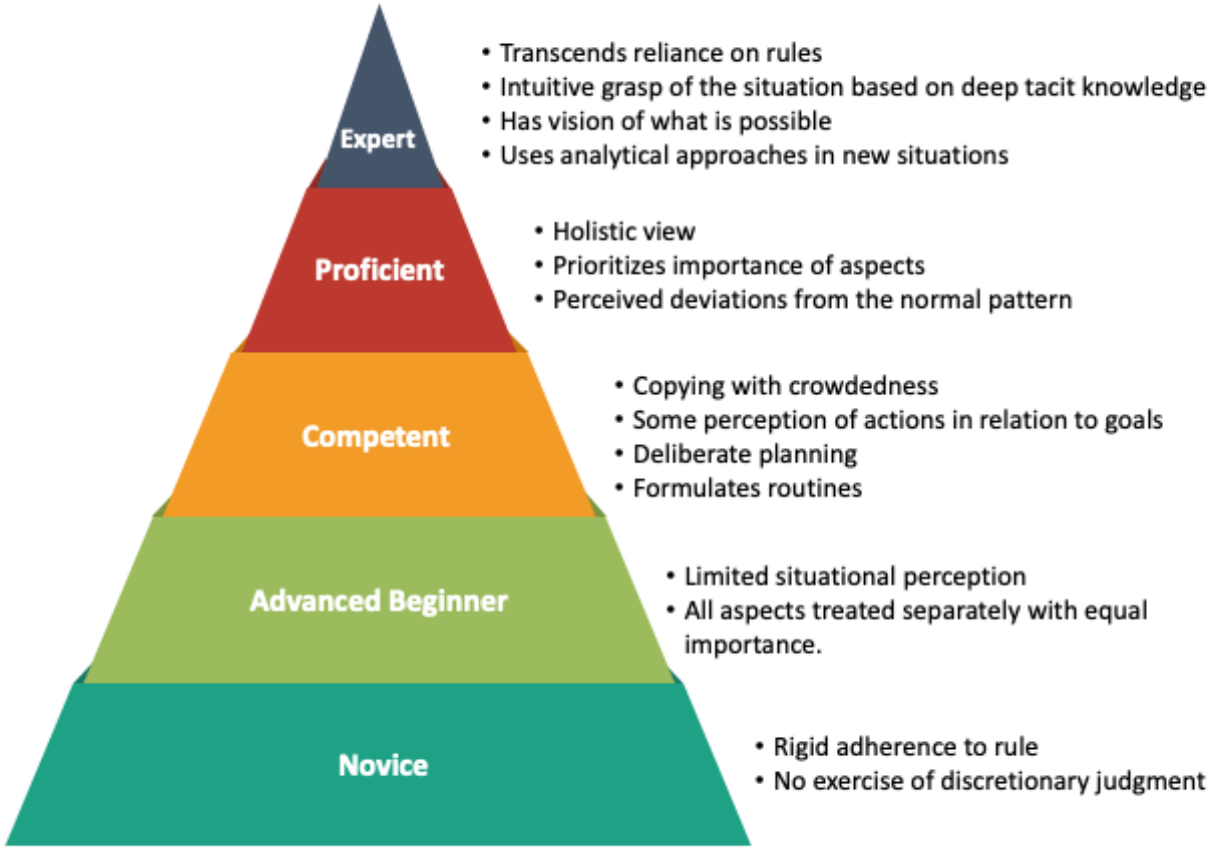
Cooperative Education (Co-op)

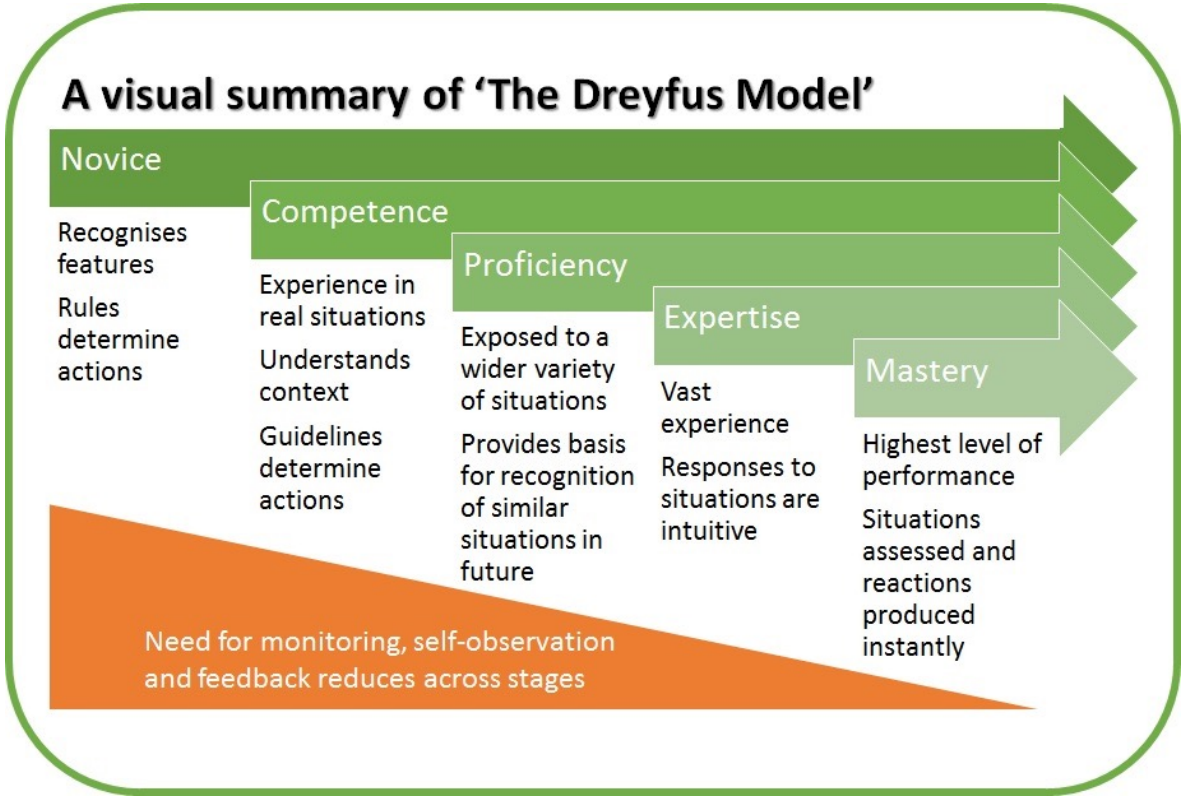
cooperative education—an educational model in which students alternate their classroom learning with sustained, full- time immersion in the professional workplace and then integrate the two.



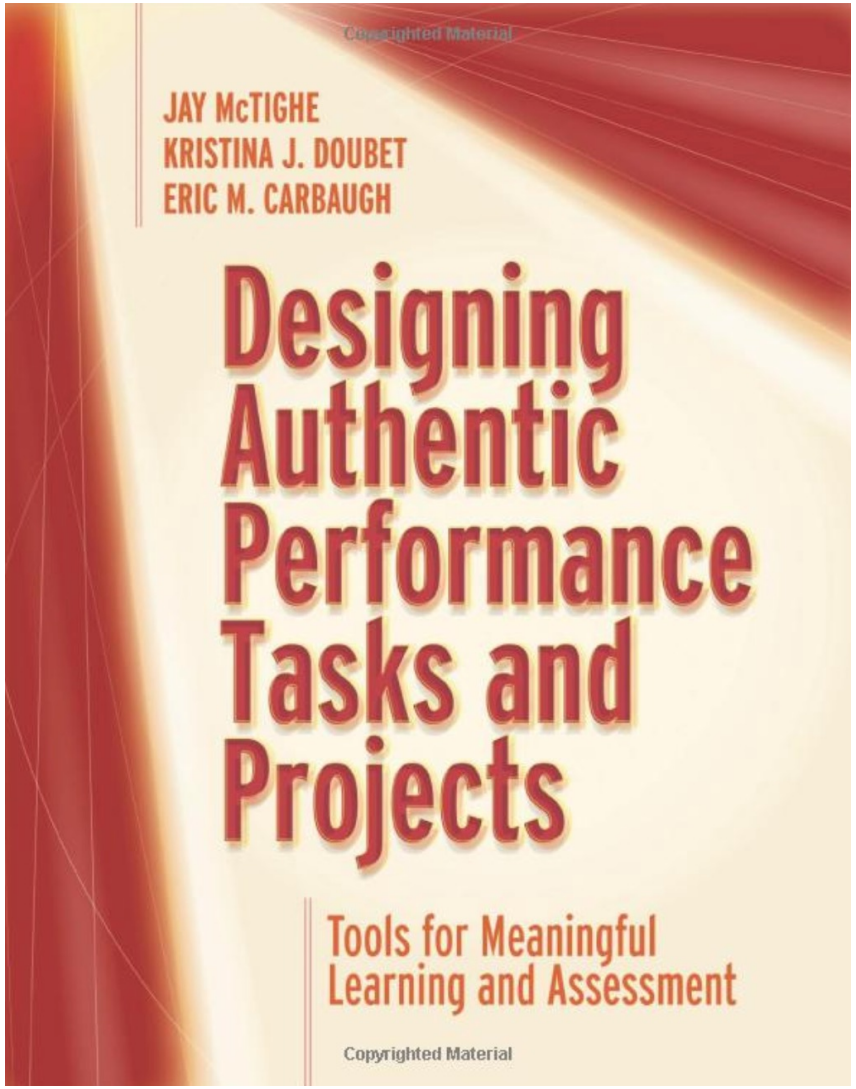
Expertise

Expertise results from acquisition, integration, and application.





Authentic Learning



Become a Coach: Provide feedback



Students learn best when they receive formative feedback about specific qualities of their work along with concrete advice on what they can do to improve.

Leveraging ChatGPT for Formative Assessment

Why assess?

(1) to assist learning; (2) to measure individual achievement; and (3) to provide accountability and to evaluate performance.

Summative Assessment



Assessment "of" learning

Formative Assessment



Assessment "for" learning

In a landmark study, Black and Wiliam showed that formative assessments, in contrast to summative ones, return the more powerful effect on student learning.